

Strict dilation approximants for the BAP question in holomorphic-Lipschitz linearization

arXiv automatic math run `fa_banach_001`, lane 10

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Status

This packet records a likely valid partial result related to Question 1 of Aron–Dimant–Garcia-Lirola–Maestre [1]. The source asks whether $\mathcal{G}_0(B_X)$ has the bounded approximation property whenever X has the bounded approximation property. The result below does not settle the endpoint identity operator. It proves that the natural dilation operators

$$D_r \delta(x) = \delta(rx)$$

are bounded-strong limits of finite-rank operators whenever r lies strictly below the reciprocal of a BAP constant for X . Thus the source’s scaling obstruction is sharp at the level of strict dilations.

Source question

The question appears after the source proof that X has the MAP if and only if $\mathcal{G}_0(B_X)$ has the MAP. The authors note that their MAP argument does not adapt directly to BAP because finite-rank approximations of the identity may send B_X into a larger ball.

$\mathcal{G}_0(B_X)$ has the MAP.

□

Note that our arguments cannot be adapted to the case in which X has the BAP since the approximations of the identity could send the unit ball B_X to a bigger ball (and, hence, we cannot control the Lipschitz norm of $P \circ T_\alpha|_{B_X}$ as in Proposition 3.1 (b)).

Question 1. Does $\mathcal{G}_0(B_X)$ have the BAP whenever X has the BAP?

The same question for $\mathcal{G}^\infty(B_X)$ was posed by Mujica in [36]. As far we know, this question is still open.

Figure 1: Question 1 on page 12 of arXiv:2304.07149, together with the scaling obstruction immediately preceding it.

Definitions and notation

Let X be a complex Banach space and B_X its open unit ball. The space $\mathcal{HL}_0(B_X)$ consists of scalar holomorphic Lipschitz functions on B_X vanishing at 0, with norm equal to the Lipschitz constant.

Its canonical predual is denoted $\mathcal{G}_0(B_X)$, and $\delta : B_X \rightarrow \mathcal{G}_0(B_X)$ is the evaluation map. The source paper proves the linearization property: every $\phi \in \mathcal{H}L_0(B_X, Y)$ has a unique bounded operator

$$T_\phi : \mathcal{G}_0(B_X) \rightarrow Y, \quad T_\phi \delta(x) = \phi(x),$$

with $\|T_\phi\| = L(\phi)$.

If X has λ -BAP, we fix a net (A_α) of finite-rank operators on X such that $\|A_\alpha\| \leq \lambda$ and $A_\alpha x \rightarrow x$ for each $x \in X$.

Proof intuition

The source MAP proof fails for BAP because $A_\alpha(B_X)$ may leave B_X . If we first shrink by a factor $r < 1/\lambda$, however, $rA_\alpha(B_X) \subset B_X$, so the holomorphic-Lipschitz linearization is available. The resulting operator need not be finite rank, since even a finite-dimensional ball has an infinite-dimensional holomorphic-Lipschitz free space. But it factors through $\mathcal{G}_0(B_E)$, where $E = A_\alpha X$ is finite-dimensional; $\mathcal{G}_0(B_E)$ has the MAP by the source theorem. Replacing the middle identity by finite-rank MAP approximants makes the factorization finite rank without changing the uniform bound. Letting $A_\alpha \rightarrow I$ then gives the strict dilation D_r .

The partial theorem

Theorem 1. *Let X be a complex Banach space with λ -BAP. For every $0 < r < 1/\lambda$, define $D_r : \mathcal{G}_0(B_X) \rightarrow \mathcal{G}_0(B_X)$ by*

$$D_r(\delta(x)) = \delta(rx), \quad x \in B_X.$$

Then D_r belongs to the strong operator closure of the finite-rank operators on $\mathcal{G}_0(B_X)$ of norm at most $r\lambda$. In particular, every strict dilation below the BAP radius has a uniformly bounded finite-rank approximation scheme on $\mathcal{G}_0(B_X)$.

Proof. Fix r with $0 < r < 1/\lambda$. Let $(A_\alpha) \subset \mathcal{L}(X, X)$ be finite-rank, $\|A_\alpha\| \leq \lambda$, and $A_\alpha x \rightarrow x$ pointwise. Put $E_\alpha = A_\alpha X$, equipped with the inherited norm from X .

The map

$$\phi_\alpha : B_X \rightarrow B_{E_\alpha}, \quad \phi_\alpha(x) = rA_\alpha x$$

is holomorphic, fixes 0, and is Lipschitz with

$$L(\phi_\alpha) \leq r\lambda < 1.$$

By linearization it induces

$$U_\alpha : \mathcal{G}_0(B_X) \rightarrow \mathcal{G}_0(B_{E_\alpha}), \quad U_\alpha \delta_X(x) = \delta_{E_\alpha}(rA_\alpha x),$$

with $\|U_\alpha\| \leq r\lambda$.

Let $j_\alpha : E_\alpha \hookrightarrow X$ be the inclusion. Since $\|j_\alpha\| = 1$, the canonical composition/linearization map

$$J_\alpha : \mathcal{G}_0(B_{E_\alpha}) \rightarrow \mathcal{G}_0(B_X), \quad J_\alpha \delta_{E_\alpha}(y) = \delta_X(y),$$

has norm at most 1. Hence

$$S_\alpha := J_\alpha U_\alpha$$

satisfies $\|S_\alpha\| \leq r\lambda$ and

$$S_\alpha \delta_X(x) = \delta_X(rA_\alpha x).$$

For each α , the space E_α is finite-dimensional, hence has the MAP. By Theorem 3.2 of [1], $\mathcal{G}_0(B_{E_\alpha})$ has the MAP. Therefore there is a net $(R_{\alpha,\beta})_\beta$ of finite-rank operators on $\mathcal{G}_0(B_{E_\alpha})$, with $\|R_{\alpha,\beta}\| \leq 1$, converging strongly to the identity on $\mathcal{G}_0(B_{E_\alpha})$. The operators

$$F_{\alpha,\beta} := J_\alpha R_{\alpha,\beta} U_\alpha : \mathcal{G}_0(B_X) \rightarrow \mathcal{G}_0(B_X)$$

are finite rank and satisfy

$$\|F_{\alpha,\beta}\| \leq r\lambda.$$

For fixed α , they converge strongly to S_α .

Finally, S_α converges strongly to D_r . Indeed, for each $x \in B_X$,

$$S_\alpha \delta_X(x) = \delta_X(r A_\alpha x) \longrightarrow \delta_X(rx) = D_r \delta_X(x),$$

because δ_X is Lipschitz and $A_\alpha x \rightarrow x$. The operators (S_α) are uniformly bounded by $r\lambda$, and $\text{span}\{\delta_X(x) : x \in B_X\}$ is dense in $\mathcal{G}_0(B_X)$, so the convergence holds on all of $\mathcal{G}_0(B_X)$.

Combining the two strong-closure steps, D_r lies in the strong operator closure of finite-rank operators of norm at most $r\lambda$. $\square$

Remark 1. *If X has the MAP, one may take $\lambda = 1$ and let $r \uparrow 1$, which recovers the endpoint approximation of the identity. For a genuine $\lambda > 1$ BAP approximation scheme, the theorem stops at $r < 1/\lambda$; the missing step is precisely a uniform way to cross the boundary from strict dilations to $D_1 = \text{Id}$.*

Verification and novelty checks

No numerical computation is involved. The proof uses only the source linearization theorem, the source MAP equivalence for finite-dimensional spaces, and the definition of λ -BAP.

Local duplicate checks found an earlier 2304.07149 scaling-obstruction attempt, which recorded the obstruction but did not promote a theorem. Web searches on June 18, 2026 for the exact title, for $\mathcal{G}_0(B_X)$ with BAP, and for holomorphic Lipschitz bounded approximation found the source arXiv paper and the related 2026 composition-operator paper, but no full answer to Question 1. The 2026 paper uses BAP for composition-operator characterizations; it does not appear to settle the BAP of $\mathcal{G}_0(B_X)$.

Human-review recommendation

Review as a modest but clean partial theorem. The main point to check is the factorization

$$\mathcal{G}_0(B_X) \xrightarrow{U_\alpha} \mathcal{G}_0(B_{E_\alpha}) \xrightarrow{J_\alpha} \mathcal{G}_0(B_X)$$

and the use of the MAP of $\mathcal{G}_0(B_{E_\alpha})$ to replace the middle identity by finite-rank maps while preserving the $r\lambda$ bound.

References

- [1] R. Aron, V. Dimant, L. C. Garcia-Lirola, and M. Maestre, *Linearization of holomorphic Lipschitz functions*, arXiv:2304.07149, 2023.